Sampling & Quantization

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Introduction

- Sampling and quantization are fundamental processes in converting analog signals into digital form.
- These processes are essential for digital communication, storage, and processing of signals.
- Understanding their problems and effects helps improve signal quality and system performance.

01

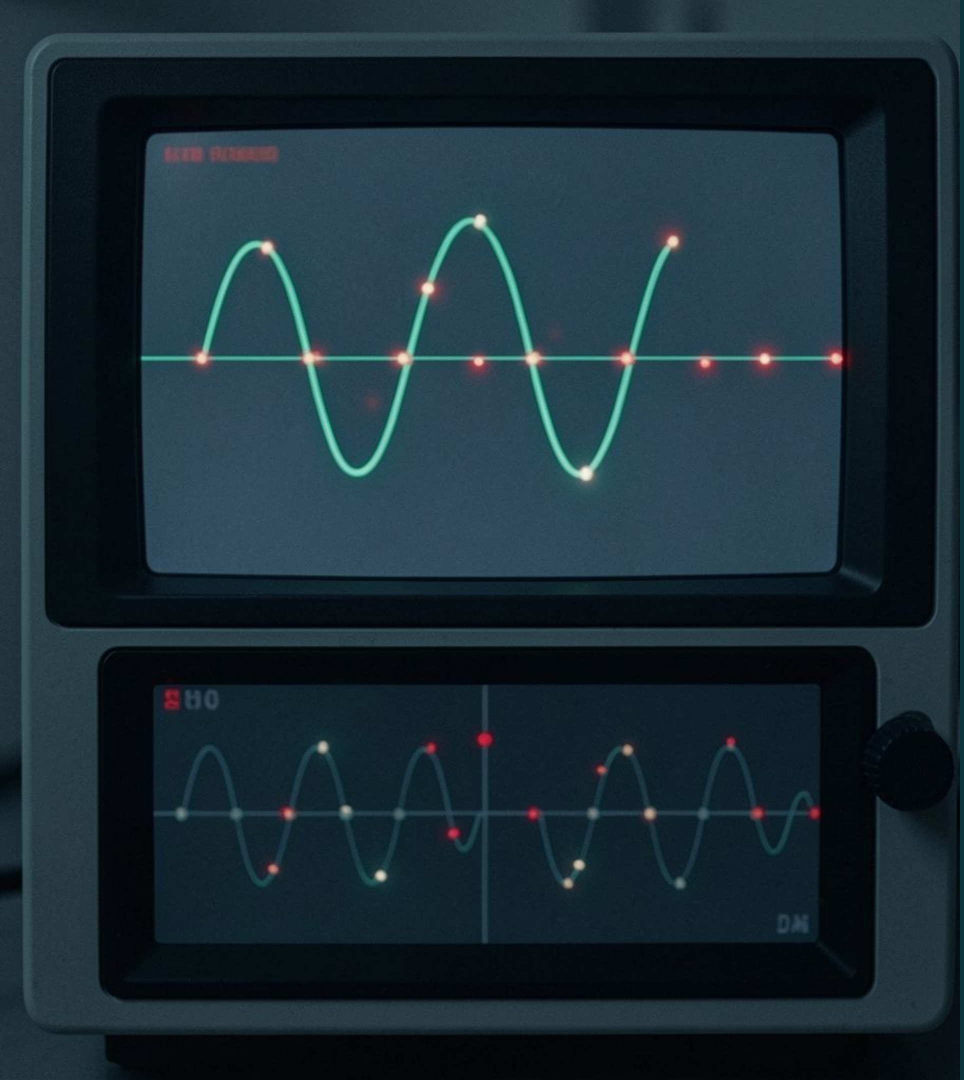
Sampling Theory

Nyquist and aliasing

- Sampling must exceed twice the highest signal frequency (*Nyquist rate*) to avoid **aliasing**.
- When rate is insufficient, high-frequency components fold into lower bands—visible in time and frequency plots. Use anti-aliasing filters and proper sampling rates to preserve signal integrity.

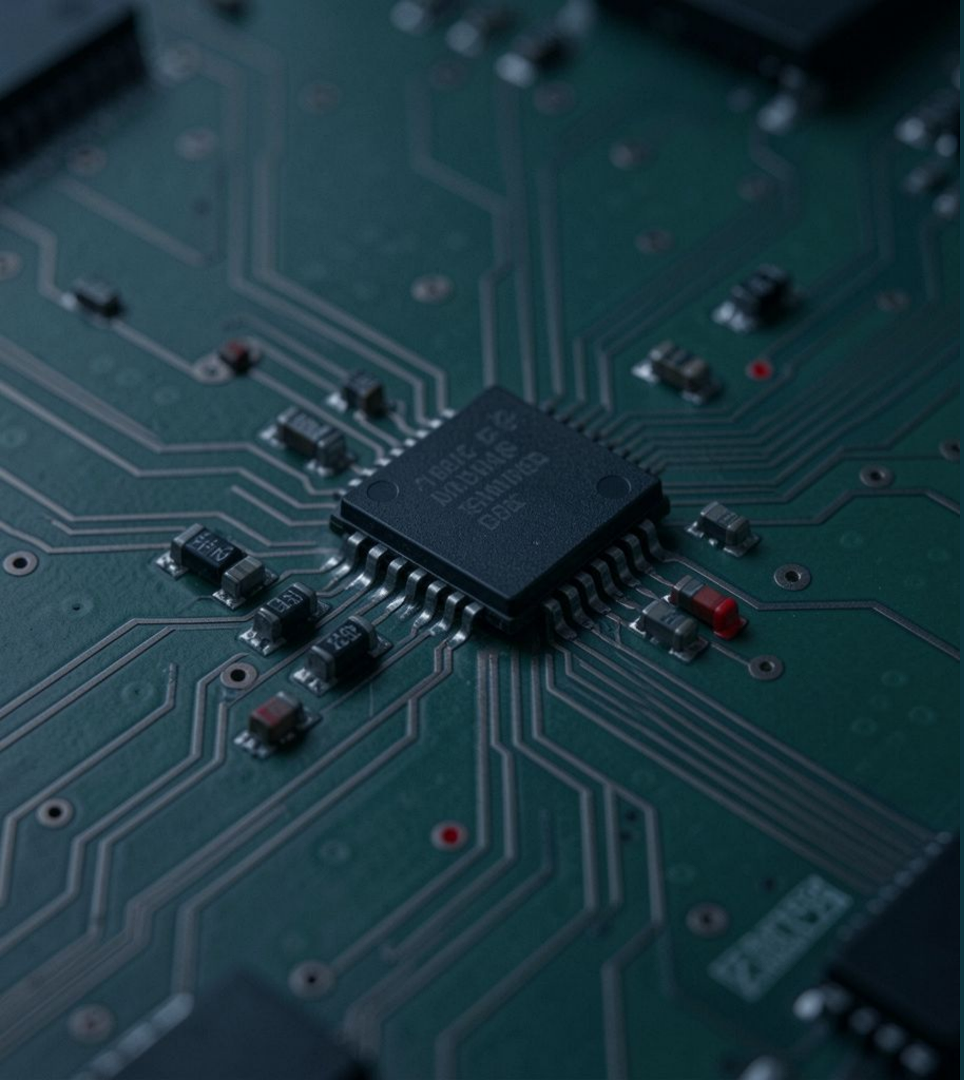
Under-sampling effects

- Under-sampling produces spectral overlap and distorted reconstructions, causing misinterpretation of signal content.
- Simulations show clean waveform at adequate rates and pronounced artifacts when under-sampled. Remedies: increase sample rate, apply analog filtering, or use controlled undersampling techniques with known band structure.



Sampling visualizations

- Visual plots compare continuous signal, sampled points, and reconstructed waveform to show effects of different rates.
- Spectral diagrams illustrate folding when below *Nyquist rate*, and clean separation when adequate. Use simulations to compare time-domain error and frequency-domain aliasing metrics.



02

Quantization Effects

Quantization fundamentals

- Quantization maps continuous amplitudes to discrete levels; resolution defined by *bit depth* and step size.

- Lower resolution increases **quantization error** and effective noise floor. Design trade-offs: bit depth vs storage, and dithering to reduce distortion.

Over-quantization & noise

- Excessive quantization (too few bits) raises noise and masks low-amplitude details, reducing dynamic range.
- Visualize histograms of quantization error and SNR degradation. Mitigation: increase bit depth, apply noise-shaping or analog preconditioning.

Quantization visual examples

- Side-by-side waveforms show original, coarsely quantized, and finely quantized signals.
- Spectrograms and error plots quantify distortion. Key takeaway: **adequate bit depth** preserves fidelity; marginal bits produce audible/visible artifacts.

Conclusion

- Proper sampling and quantization preserve signal integrity by meeting *Nyquist* and using sufficient bit depth.

- Combine anti-aliasing, appropriate sampling rates, and adequate quantization to minimize aliasing and noise. Use simulations and visual checks to validate system choices.

THANK YOU